LESSON 6.1

# Division Expressions with Fractions and Whole Numbers

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can explain what dividing with a whole number and a unit fraction means — how many groups fit, or how much is in one group — find the quotient with a model, and check it by multiplying.

**WHY THIS MATTERS** Agent Reyes finds a 3-foot strip of tape at the crime scene. The lab needs the tape cut into 1/4-foot sections for analysis. How many sections can be made? The answer cracks the code to the locked evidence vault.

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Reciprocal** *Spanish: Recíproco* | A fraction turned upside down. | 1/4 → 4/1 |
| **Quotient** *Spanish: Cociente* | The answer when you divide. | 8 ÷ 1/2 = 16 is the quotient. |
| **Dividend** *Spanish: Dividendo* | The total number being divided into equal groups (the number inside the division bracket). | In 8 ÷ 1/2, 8 is the dividend. |
| **Divisor** *Spanish: Divisor* | The number of equal groups you are dividing into (the number outside the division bracket). | In 8 ÷ 1/2, 1/2 is the divisor. |
| **Unit fraction** *Spanish: Fracción unitaria* | A fraction with 1 on top, like 1/4. | 1/3, 1/8, 1/10 |
| **Multiplicative inverse** *Spanish: Inverso multiplicativo* | Two numbers whose product is 1; each is the reciprocal of the other. | 4 and 1/4 · 2/3 and 3/2 |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**A cook has 4 cups of broth. How many 1/3-cup servings can she pour?**

1. Write the division: 4 ÷ 1/3.
2. Keep 4, change ÷ to ×, flip 1/3 to 3: 4 × 3 = 12.
3. Check: 12 × 1/3 = 4 cups.

**Answer:** 12 servings

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**Let's try 2 ÷ 1/3. The bar model counted 6 thirds inside 2 wholes — now let's see why.**

1. _____
2. _____
3. _____
4. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- To divide by a fraction, I multiply by its _____ .
- I wrote _____ over 1 so both numbers are fractions.
- The quotient is _____ because that many groups fit inside.
- Dividing by a fraction smaller than one whole makes the quotient _____ .

**Word bank** reciprocal · quotient · dividend · divisor · unit fraction · keep · change · flip · multiply · groups · fit inside · larger

WATCH OUT!

## A common mistake

A common mistake in Interpret Division of Fractions is assuming that dividing always makes the answer smaller. For 4 ÷ 1/3, students reason 'dividing shrinks the number' and compute 4 ÷ 3 ≈ 1.3 instead of asking how many 1/3-foot pieces fit into 4 feet (each whole foot holds 3 thirds, so 4 × 3 = 12). Before you submit, check: if the divisor is a fraction smaller than 1, the quotient must be BIGGER than the dividend, not smaller.

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**Level 0** · Most support · every hint printed

**MASTERY CHECK**
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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## PART 1 Understand the idea

models and meaning

**1** **Circle the letter of the best answer. Show how you know.**

What does 3 ÷ 1/4 mean?

(A) How many 1/4-size pieces fit into 3
(B) 1/4 of 3
(C) 3 groups of 1/4
(D) 3 minus 1/4

**STEPS TO TRY**

1. Re-read the expression: 3 is the total amount and 1/4 is the size of one piece. What question could that setup be asking?
2. Dividing by a fraction asks a counting question about pieces, not 'a part of.' Predict: should the result be bigger or smaller than 3?
3. Picture a 3-foot tape cut into 1/4-foot pieces, then set it up: put a 1 under the 3 and flip the 1/4.

SHOW YOUR WORK

**2** **Circle the letter of the best answer. Show how you know.**

Which expression means 'how many 1/3-size pieces are in 2'?

(A) 1/3 ÷ 2

(B) 2 ÷ 1/3

(C) 2 × 1/3

(D) 2 + 1/3

**STEPS TO TRY**

1. Find the two jobs: which number is the total being split, and which is the size of each piece?
2. Division setups read total ÷ piece size. The total always sits in front of the ÷ sign.
3. Here the total is 2 and each piece is 1/3. Build the expression with the total first, then check it against each choice.

**SHOW YOUR WORK**

## PART 2 Apply it

real-world problems

**3** **Circle the letter of the best answer. Show how you know.**

A rope is 4 feet long. How many 1/2-foot pieces can be cut from it?

(A) 1/2 piece

(B) 2 pieces

(C) 4 pieces

(D) 8 pieces

**STEPS TO TRY**

1. Note the numbers: the rope is 4 feet long, and every piece is 1/2 foot — smaller than a whole foot.
2. Start small: figure out how many 1/2-foot pieces fit into just 1 foot of rope.
3. Dividing by a fraction is the same as multiplying by its reciprocal: 4/1 × 2/1.

**Write any whole number over 1** _____

**Keep · Change · Flip** _____

**Multiply across** _____

**Simplify · label the units** _____

1. **Whole over 1** *3 becomes 3/1.*
2. **Keep** *The first fraction does not move.*
3. **Change · Flip** *÷ becomes ×; flip the second fraction.*
4. **Check** *Dividing by a piece smaller than 1 gives a BIGGER answer.*

**4** **Circle the letter of the best answer. Show how you know.**

A half-pan of brownies is shared equally among 4 people. How much of the whole pan does each get?

(A) 1/8
(B) 1/2
(C) 2
(D) 4/2

**STEPS TO TRY**

1. Draw half a pan, then split that half into 4 equal parts.
2. How many of those parts fit in the WHOLE pan?
3. The whole number is second here. Put a 1 under the 4 to get 4/1, then flip it to 1/4 and multiply.

**SHOW YOUR WORK**

**EXPLAIN YOUR THINKING**

## Why does dividing by a unit fraction always give a number larger than the dividend?

**SENTENCE STARTER** Dividing by 1/3 means finding how many ___ fit into ___, so the quotient is ___ than the dividend.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help